TRIBHI KATHURIA

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My research focuses on preference learning for robots operating in human-centered environments. I study how robots can learn socially aware navigation and planning behaviors from demonstrations and human feedback, and validate these on real robotic systems in assistive and interactive settings.

EDUCATION

University of Michigan | Ph.D. in Robotics — Sep 2020 – Dec 2024

Lab — Computational Autonomy and Robotics Lab (CURLY), PI: Maani Ghaffari Jadidi; Interaction and Collaboration Research Lab (ICRL), PI: Xi Jessie Yang

Research Focus — Preference learning for robots that work with people, through Inverse Reinforcement Learning, Learning from Demonstration, and Behavior Planning.

University of Michigan | M.S. in Robotics (GPA: 3.80/4.00) — Sep 2018 – May 2021

Coursework — Self-driving Cars, Motion Planning, Mobile Robotics (SLAM), Reinforcement Learning, Math for Robotics, Robotic Systems Lab, Nonlinear Systems and Control.

NSIT, University of Delhi | B.E. in Instrumentation and Control Engineering, India — Aug 2014 – Jun 2018

ACADEMIC APPOINTMENTS

Robotics Institute, University of Michigan | Lecturer III — Sep 2025 – Present

- Instructor for ROB 101 (Computational Linear Algebra, about 170 students per semester) and ROB 320 (Robot Operating Systems, about 30 students per semester).
- Introduced flipped-classroom-style labs for ROB 101, asking students to translate math to Julia code while talking through the problem sets.
- ROB 101 lead for the Distributed Teaching Collaborative (DTC) with Howard University, remotely supporting Howard students taking the course.
- Building on this collaboration, co-developed a proposal to extend ROB 101 to support AI education in R2 schools, presented at the AWS-MLU symposium: Link
- Developed lecture content and updated lab materials for ROB 320 for Fall 2025, supporting a project-based curriculum in which students built a ROS stack on an MBot platform and simulated a Fetch robot with correct kinematics.
- Taught Publish-Subscribe paradigm, where students used concepts from TCP communication over sockets and multi-threading to run velocity commands on an Omni-drive robot platform (Mbot).

GRANTS AND AWARDS

Emergent Ventures Grant, Mercatus Center, George Mason University | PI, $10K — 2026

- Awarded to develop a remote AI/robotics education and mentoring program connecting undergraduate and high school students in countries such as India with graduate student mentors, using a curriculum I developed.

Towner Prize for Outstanding Graduate Student Instructors | Honorable Mention — 2022

- University of Michigan. Award listing: https://crlte.engin.umich.edu/towner-prize-winners/

RESEARCH EXPERIENCE

University of Michigan | Independent Research — Sep 2025 – Present

- Developing **Hierarchical Reinforcement Learning** based navigation for dense, crowded environments, where a high-level policy switches between pretrained low-level skills using the robot's first-person and top-down images.
- Submitted a paper to ICRA 2027 presenting a hierarchical planner evaluated on a narrow crossing scenario.
- Beginning work on **preference learning** from da Vinci surgical camera data with Wayne State University.
- Mentor undergraduate students in creating datasets of socially aware scenarios in photo-realistic simulation.

University of Warwick and University of Leeds | Visiting Researcher, UK — Aug 2026

- Hosted by Prof. Emma MacPherson (University of Warwick) and Prof. Pietro Valdastri (University of Leeds); conducted robotic THz experiments at Warwick.
- Developed a **Gaussian process regression** model that estimates probe-skin contact quality from force-sensor feedback and the THz sensor posterior.
- Used the model to correct probe contact during robotic THz scanning, enabling reliable readings and reducing clinical scan time.
- Submitted a paper to ICRA 2027 on sparse contact prediction and probe correction for robotic THz scanning.
- Continuing the work remotely, now dynamically closing contact gaps by moving the robot with the scanning stage.

University of Michigan | Graduate Student Research Assistant Sep 2020 – Dec 2025

- Developed motion and behavior planning algorithms for Socially Assistive Robots navigating dense, dynamic indoor environments.
- Developed an **Inverse Reinforcement Learning** based local planner to learn robot navigation behaviors from raw sensor data. Presented results on **crowd navigation** and **deadlock avoidance** for robots navigating around humans.
- Incorporated reasoning from **foundation models** to differentiate good vs. bad social behavior and supervise the ranking of demonstration data.
- Designed a **Mixed Integer Linear Programming** based tour planner that handles multiple visitor constraints to plan routes on a spatial-semantic graph, used as the global planner for an assistive robot.

Perceptual Robotics Lab (PeRL), University of Michigan | Research Assistant Sep 2019 – Sep 2020

- Implemented an MDP planner for navigating a museum with a Fetch robot, tested in simulation in Gazebo.
- Implemented **Augmented MDP** and **MC-POMDP** (approximate POMDP) planners on a simulated grid-world problem.

Honda R&D Americas | Automated Vehicle Control Prototyping Intern, Ann Arbor, MI May 2019 – Dec 2019

- Developed a control algorithm to carry out the longitudinal control of the vehicle for a custom application. Communicated with a DSRC roadside unit (V2I) to get information about the remote vehicles to the host vehicle for decision making and control.
- Controlled the speed and the brake of the vehicle, communicating messages to the Vehicle CAN via a MicroAutobox (RCP).
- Planned the trajectory of the vehicle for L2 autonomy using a pure pursuit algorithm with GPS data for feedback. Successfully completed real-world testing of the application for a company showcase.

PROJECTS

Robot Guide at the University of Michigan Museum of Art Jun 2022 – Aug 2024

- Used a **GPT davinci-002 model** agent to fine-tune responses from the robot to give specific answers about the artworks at the museum, including contextual prompts like "Tell me more about this?", with ROS support. Code available at: https://github.com/UMich-CURLY/Hi-Fetch
- Built an offline map of the museum and localized online to plan and complete a tour with small groups of participants walking behind the robot. The agent introduced the artwork and then could handle follow-up questions.
- Mentored an iOS app development project that let the robot showcase museum exhibits and introduce them to a visiting class of high school students.
- Demo video available at: https://youtu.be/SgwyzEO0flY?si=7JQ-e9seXoiLh7Ym

Motion Planning under Uncertainty Jan 2019 – Mar 2019

- Implemented discrete (A*, ANA*) and sampling-based (RRT, Bi-RRT) motion planners for the PR2 robot in a cluttered environment in OpenRAVE.
- Proposed a particle RRT where the different particles were used to characterize the uncertainty in the robot model and do re-planning when the maximum likelihood particle trajectory was far from the planned trajectory.

Robotic Systems Lab, University of Michigan Sep 2018 – Dec 2018

- Balancebot Project: Designed a PID-controller for the balancing motion of a two-wheeled robot and drove it autonomously in a maze captured by a motion capture system on a path planned using a Virtual Potential Field.
- Mobilebot Project: Performed Monte-Carlo localization for a LiDAR equipped mobile robot and used A* path planning for optimal exploration to map an unknown environment.

PUBLICATIONS

- X. Wang, **T. Kathuria**, **(Under Review)** Hierarchical Policy Learning for Socially Aware Robot Navigation. *2027 IEEE International Conference on Robotics and Automation (ICRA).*
- X. Gui, **T. Kathuria** et al., **(Under Review)** Sparse Contact Prediction and Probe Correction for Robotic Terahertz Scanning. *2027 IEEE International Conference on Robotics and Automation (ICRA).*
- **T. Kathuria** et al., **(2025)** Learning Implicit Social Navigation Behavior using Deep Inverse Reinforcement Learning. *IEEE Robotics and Automation Letters (RA-L).* IEEE Xplore
- Y. Xu, T. Chakhachiro, **T. Kathuria** et al., **(2023)** SoLo T-DIRL: Socially-Aware Dynamic Local Planner based on Trajectory-Ranked Deep Inverse Reinforcement Learning. *2023 IEEE International Conference on Robotics and Automation (ICRA).* DOI: 10.1109/ICRA48891.2023.10160536
- **T. Kathuria** et al., **(2022)** Providers-Clients-Robots: Framework for spatial-semantic planning for shared understanding in human-robot interaction. *31st IEEE International Conference on Robot & Human Interactive Communication (ROMAN).* DOI: 10.1109/RO-MAN53752.2022.9900525
- B. Fu, **T. Kathuria** et al., **(2021)** Simultaneous Human-Robot Matching and Routing for Multi-robot Tour Guiding under Time Uncertainty. *Journal of Autonomous Vehicles and Systems Special Issue.* DOI: 10.1115/1.4053428

- **T. Kathuria** et al., **(2018)** Control of Three Link Manipulator using Fractional Order PID Controller. *Fractional Order Systems, 2018 - Elsevier.* DOI: 10.1016/B978-0-12-816152-4.00016-9
- **T. Kathuria** et al., **(2017)** Study of Optimization Methods for Tuning of PID Gains for Three Link Manipulator. *Proc. of 7th International Conference on Cloud Computing, Data Science & Engineering - Confluence.* DOI: 10.1109/CONFLUENCE.2017.7943131

PRESENTATIONS

- ICRA 2023 Workshop L3H2 - Lifelong Learning with Human Help. Presented work title: Human-Aware motion planning using IRL on spatial semantic graphs. https://life-long-learning-with-human-help-l3h2.github.io
- ICRA 2022 Workshop ROPM - Robotic Perception And Mapping . https://sites.google.com/view/ropm/home
- ROMAN 2022 Talk - Providers-Clients-Robots: Framework for spatial-semantic planning for shared understanding in human-robot interaction. https://doi.org/10.48448/eh1p-8060
- COMCO 2021 - Computational Cognition Conference. Planning architecture for SARs (Poster). https://comco2021.uni-osnabrueck.de/abstracts/day1_poster_kathuria

PREVIOUS TEACHING EXPERIENCE

University of Michigan | Graduate Student Instructor, Self-Driving Cars — Sep 2023 – Dec 2023
- Held regular office hours, with emphasis on control theory.
- Gave a guest lecture on task planning and routing for self-driving vehicles, with hands-on examples in Python.
- Homeworks and content available at: https://github.com/UMich-CURLY-teaching/UMich-NA-565.git

University of Michigan | Graduate Student Instructor, Computational Linear Algebra — Sep 2020 – Dec 2021
- Co-created content for the Pilot offering of a Linear Algebra course for Freshmen, including lab work and recitation material. Led the LiDAR project development.
- Led regular recitations. Videos and content: https://github.com/michiganrobotics/rob101

University of Michigan | Guest Lecturer, Mobile Robotics — Apr 2019
- Lecture on Motion Planning for Autonomous Systems. Introduction to Reinforcement learning, and background with MDP and POMDPs.

PROFESSIONAL SERVICE

Departmental Service
- Member, Undergraduate Curriculum Committee (UGCOM), University of Michigan Robotics — 2025–Present
- Member, Undergraduate Awards Committee, University of Michigan Robotics — 2025–Present
- Faculty Advisor, Robotics Undergraduate Student Committee (RUSC) — 2025–Present
- Qualifying Exam (CQE) Examiner, University of Michigan Robotics — 2025–2026
- Participated in faculty hiring discussions

Peer Reviewing
- IEEE Transactions on Robotics (T-RO) - 2024, 2025, 2026
- IEEE Transactions on Automation Science and Engineering (T-ASE) - 2026
- IEEE Robotics and Automation Letters (RA-L) - 2022, 2024, 2025, 2026
- IEEE International Conference on Robotics and Automation (ICRA) - 2020, 2022, 2023, 2024, 2025
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) - 2020, 2022, 2023, 2024, 2025, 2026
- International Conference on Ubiquitous Robots (UR) - 2020

Mentoring
- Xinyuan Wang (Hierarchical Policy Learning for Social Navigation) — Current: Undergraduate Student, UMich
- Xiangyu Guo (Robotic THz Scanning) — Current: Ph.D. Student, University of Warwick
- Yifan Xu (Global Task/Tour Planning) — Next Position: Ph.D. Student, UMich
- Theodor Chakhachiro (Global Tour/Task Planning) — Next Position: Controls Engineer, Figure AI
- Ke Liu (Navigating narrow environments) — Next Position: Software Engineer, ByteDance/TikTok
- Erik Chi (Tour Guide Demo) — Next Position: Ph.D. Student, UMich

Outreach
- Shared Robotics resources and experiences in GuruTalks: https://youtu.be/hKLp_sW0t9A — 2022
- Mentoring undergraduate DEI scholars — 2021, 2022
- Student Council Social Chair — 2021
- Member of Robotics DEI committee — 2021

· Mentor for UMich Robotics Mentorship Program 2019

TECHNICAL SKILLS

· **Machine Learning Tools:** Pytorch, Tensorflow, Open AI Gym, CUDA, Tensorboard, Visdom.
· **Robotics and Simulation:** ROS, ROS Navigation Stack, Gazebo, lcm, OpenRAVE, CANalyzer, LABVIEW.
· **Programming Languages:** C++, C, Python, Julia, MATLAB/Simulink, ControlDesk.
· **Tools/Frameworks:** Linux, git, WSL, Docker, Jupyter, Conda.